

Abhinav Agrawal

Bengaluru, Karnataka, India

Email: abhinav.hfs@gmail.com | Phone: +91 8169809401 | LinkedIn: LinkedIn | GitHub: skully-coder

Summary

Senior Applications Engineer in Oracle Transportation Management (OTM) with 2+ years of experience building enterprise-grade optimization and algorithmic capabilities in the Transportation & Global Trade Management domain. Focused on mixed-integer programming (MIP), sequencing/packing optimization, and applying GenAI/agentic workflows to improve product usability, reliability, and operational outcomes.

Core Skills

Algorithms & Optimization (MIP, Packing, Sequencing) | Java | Python | C/C++ | REST APIs & Microservices | GenAI & Agentic Workflows (LLM-based apps, multi-agent concepts) | Cloud & Distributed Systems | Agile/Scrum | JIRA

Experience

Oracle — Bengaluru, India

Senior Applications Engineer

Sep 2025 – Present

Oracle Transportation Management (Algorithms / Optimization)

- Developed a novel mixed-integer programming (MIP)–based packing optimization approach for bulk products, improving solution efficiency by 35% and reducing execution time by 50%.
- Built a agentic AI application by integrating Oracle Transportation Management (OTM) and Oracle Warehouse Management (WMS) using Fusion AI App Builder to detect supply-chain disruptions, alert managers, and automatically recommend corrective actions.
- Driving adoption of GenAI and agentic AI workflows within the Transportation Management product, translating new capabilities into practical, scalable features.

Application Developer

Jul 2023 – Sep 2025

Oracle Transportation Management (Algorithms / Optimization)

- Algorithms engineer on Oracle Transportation Management (OTM), driving optimization and core algorithm development.
- Delivered product enhancements that introduced new capabilities and improved existing functionality across OTM components.
- Rapidly triaged and resolved critical internal issues within committed deadlines, improving stability, reliability, and release readiness.
- Partnered directly with customers to understand implementation blockers, accelerate root-cause analysis, and recommend practical solutions to improve adoption and operational efficiency.
- Created a proof-of-concept GenAI agent that infers product documentation to generate clearer, more accurate, and more consistent answers across multiple OTM components.

Samsung R&D Institute India — Bengaluru, India

Software Engineering Intern

Mar 2022 – Aug 2022

ONNX / Model Tooling (C/C++)

- Developed C/C++ utilities to programmatically inspect, modify, and summarize ONNX models for machine learning and deep learning workflows.
- Evaluated and identified reliable conversion paths to translate models from common source frameworks

into ONNX format.

- Implemented reusable helper functions to read, validate, and update ONNX model metadata and to edit properties of individual graph nodes.
- Built APIs to modify existing node attributes and parameters safely and consistently across model graphs.
- Implemented model summarization capabilities to report graph structure, key attributes, and per-node properties to support debugging and analysis.
- Produced an Android-compatible shared library by cross-compiling for the arm64-v8a ABI, enabling on-device model inspection and transformation.

Patents

- **MIP-Based Packing for Bulk Container Optimization** — Patent Application — **Primary Inventor**. Mixed-integer programming approach for bulk container/compartment packing with constraints (capacity/compatibility/mixing rules) and split-penalty tuning to generate an executable packing blueprint/manifest.
- **Agentic Simulated Annealing with Generational Iterations** — Patent Application — Agent-population approach using simulated annealing with generational (genetic) iterations to explore complex, multi-solution problem spaces and identify qualified solutions.

Education

Manipal Institute of Technology, B.Tech. in Computer Science and Engineering Jul 2019 – Jun 2023
CGPA: 8.9/10.0

Projects

ShikshaAI — AI-enabled teaching assistant for educators in low-resource environments; automates lesson planning and creation of custom materials through a simple interface.

Cupidity — Music-matching app built using Spotify API, Node.js, MongoDB, Firebase, Flutter, and Google Cloud.

Leadership & Volunteering

- Volunteered as a coach and referee for young underprivileged football players as part of Oracle volunteering initiatives.
- Developer Student Clubs (MIT Manipal): App Dev Subhead (Sep 2021 – Aug 2022).
- IECSE Manipal: Managing Committee Member (Aug 2019 – Jul 2021).

Certifications

- Professional Scrum Master I (PSM I)
- Build and Secure Networks in Google Cloud
- Engineer Data in Google Cloud
- Integrate with Machine Learning APIs
- Global Tech Program Alum

Technologies

Programming: Java (EE/SE), Python, C, C++, JavaScript

AI/ML: TensorFlow, Keras, Pandas, Jupyter, LangChain, LangGraph, Multi-Agent Architecture

Web/Mobile: REST APIs, Microservices, Node.js, Express.js, Django, Flutter, Dart, HTML/CSS

Databases: Oracle, MongoDB, Vector DB exposure (e.g., ChromaDB)

Tools/Process: Agile, Scrum, JIRA